



Measurement-Driven Low-Label Adaptation for Signal-Based Areal Surface Roughness Prediction After Tool Replacement in Ti–6Al–4V Milling

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Abstract

In precision milling, areal surface roughness (S_a) is commonly verified through offline surface topography measurement. However, tool replacement can change process signals and reduce the reliability of signal-based S_a prediction. This study formulates S_a prediction in Ti–6Al–4V milling as a measurement-driven low-label target-tool adaptation problem. Three tools were machined under fixed cutting parameters, and each steady-state pass was paired with one confocal white-light S_a measurement. Force and vibration signals were organized using a one-pass–one-sample strategy to avoid overlapping window sample inflation. Seventy-two statistical descriptors were extracted to describe amplitude, energy, fluctuation, distribution, and spectral response characteristics. Descriptor-based regressors and a hierarchical signal statistical fusion model were evaluated using the same leave-one-tool-out (LOTO) protocol. Unlabeled target-tool descriptors were used for per-tool input normalization, and limited target-tool S_a labels were used only for affine output calibration. Raw cross-tool prediction remained unreliable for all models, with negative average R^2 values. After calibration, random forest achieved the best numerically calibrated performance in the present controlled setting, with average RMSE/ R^2 values of 0.0925 μm /0.4658, 0.0956 μm /0.3331, and 0.0973 μm /0.2908 under pass-uniform calibration, layer-all-pass calibration, and layer-one-pass calibration, respectively. Under the present controlled three-tool LOTO setting, the results indicate that reliable statistical descriptors, unsupervised input alignment, and few-shot affine correction provide a practical route for improving S_a prediction after tool replacement.

Highlights

1. Tool replacement causes severe S_a prediction bias despite fixed cutting parameters.
2. One-pass samples link steady-state signals with measured S_a values.
3. Few-shot affine calibration improves cross-tool S_a prediction with limited labels.

Keywords Ti–6Al–4V milling · Areal surface roughness · Surface metrology · Tool replacement · Statistical descriptors · Affine calibration

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1 Introduction

Surface topography metrology plays an essential role in precision manufacturing because the measured surface state directly affects friction, wear, fatigue resistance, sealing performance, and assembly contact quality [1, 2]. Areal surface roughness (S_a), provides a three-dimensional description of machined surface texture, making it more suitable

than a single profile-based index for evaluating the quality of milled surfaces. Ultimately, the cutting process is highly nonlinear and unstable due to milling of Ti–6Al–4V, low thermal conductivity, high chemical activity, and severe thermomechanical coupling [3]. Progressive tool wear further changes the tool–chip–workpiece contact state and cutting dynamics, which may alter the machined surface topography and the process signals acquired during cutting. Consequently, reliable S_a prediction after tool replacement is important for surface-quality consistency and measurement-driven process control.

In practical machining, S_a is usually obtained through offline contact or optical surface topography measurement. These measurement methods provide reliable roughness values, but they require additional measurement planning, machine stoppage, and dedicated instruments. For complex surfaces or high-throughput production, measuring every machined pass at a sufficiently high frequency can be difficult. Therefore, signal-based prediction provides a useful complement to surface metrology: force and vibration signals collected during cutting can be used to estimate the measured S_a value and support timely quality assessment before full offline inspection is completed [4]. In this sense, the prediction task is not merely a data regression problem but also a measurement-driven process-quality estimation problem.

Surface roughness prediction methods can be broadly divided into physics-based modeling, traditional machine learning, signal-feature-based modeling, and deep representation learning. Physics-based models are usually established from geometric kinematics, material-removal mechanisms, and dynamic cutting responses. Moreover, they provide clear physical interpretability. However, their accuracy may be affected by idealized assumptions, parameter uncertainty, boundary conditions, tool runout, and transient disturbances in real machining [5]. Therefore, traditional machine learning methods have been widely used to learn nonlinear mappings from machining variables to roughness. For example, Zhou et al. [6] combined genetic optimization with gradient boosting regression trees, Chen et al. [7] introduced cutting forces and vibrations into a nested artificial neural network (ANN) framework, Lu [8] reviewed surface quality prediction in terms of process parameter effects, and Wu and Lei [9] used vibration signal analysis with an ANN for milling roughness prediction. These studies demonstrated the feasibility of data-driven roughness prediction. However, models based mainly on nominal process parameters or low-dimensional handcrafted variables may be insufficient for describing real-time cutting-state variation [10].

With advances in sensing and data acquisition technologies, dynamic cutting signals such as force, vibration, acoustic emission, and temperature have been further integrated into roughness prediction models. Researchers have

extracted time-domain, frequency-domain, and time–frequency-domain features to characterize machining-state variation more directly [9, 11, 12]. These approaches improve the representation of cutting dynamics, but their performance may still greatly depend on manually designed signal processing and feature extraction procedures. When cutting conditions, noise patterns, or data distributions change, the stability and transferability of handcrafted features may deteriorate [13]. To reduce dependence on feature engineering, deep learning models have also been introduced for roughness prediction from process signals. Convolutional and sequence-aware models have shown potential in learning latent representations from acoustic emission or vibration signals [14, 15], while long short-term memory (LSTM)- and convolutional neural network (CNN)-based structures have been used to model temporal signal evolution in machining-related prediction tasks [16, 17]. However, increasing model complexity alone does not necessarily guarantee stable prediction when the tool state or data distribution changes.

More recently, structural fusion, multi-task learning, transfer learning, and wear-aware modeling have been explored to improve model usability under complex machining distributions. Cheng et al. [18] developed a multi-task learning model for simultaneous estimation of surface quality and tool wear. Wang et al. [12] incorporated tool wear information into a stacked autoencoder (SAE)-LSTM-based online roughness prediction framework, and Cooper et al. [19] used generative adversarial network (GAN)-synthesized power signals as process signatures to support surface roughness prediction. In the broader field of tool state perception and transfer modeling, Qiang et al. [20, 21] investigated cross-condition transfer learning for tool health monitoring and tool wear prediction, Shi et al. [22] incorporated wear information into cutting-force prediction, Ma et al. [23] proposed an end-to-end deep learning method for tool-wear monitoring, Li et al. [24] introduced a condition-adaptive framework for multi-condition tool wear prediction, and Fan et al. [25] analyzed tool-state-related deformation from a mechanics perspective. These studies suggest that the tool state is an important source of process variation and can reshape the mapping between cutting signals and machining responses.

Nevertheless, reliable S_a prediction after tool replacement remains insufficiently addressed from a measurement adaptation perspective. Many roughness prediction studies have implicitly assumed that training and test samples are drawn from the same or similar distributions. In real production, however, a newly installed tool may produce different force/vibration amplitudes, spectral responses, and output roughness distributions even when the cutting parameters remain unchanged. Consequently, a model learned from source tools may suffer from input distribution shifts, output mean

bias, scale mismatches, and goodness-of-fit degradation when applied to a target tool [20]. Since repeatedly collecting many surface topography measurements for every new tool is usually unrealistic, a more practical strategy is to use unlabeled target-tool signal/statistical distributions for input-level alignment and only a few measured target-tool S_a values for output-level correction [21].

To address this problem, this study constructs a pass-level S_a prediction framework that links steady-state force/vibration signals with measured S_a values. Descriptor-based regressors, including support vector regression (SVR), random forest (RF), Gaussian process regression (GPR), and least-squares boosting (LSBoost) are compared with a hierarchical signal statistical fusion model under identical leave-one-tool-out (LOTO), per-tool unsupervised normalization, and affine-calibration protocols. This fusion model serves as a structured comparison baseline instead of a predefined optimal solution. This design enables the study to determine whether compact statistical descriptors and conventional regressors offer a more reliable approach than increasing model complexity under the present small-sample, measurement-limited cross-tool setting [20, 21].

This study takes a different approach compared with general transfer learning and domain adaptation algorithms. Instead of focusing on a generic feature alignment model, this work formulates tool replacement as a measurement-driven, low-label target-tool adaptation problem for pass-level S_a prediction. The target response is the measured surface topography parameter S_a , and each valid steady-state cutting pass is paired with one measured S_a value. Under this setting, unlabeled target-tool signal/statistical descriptors are used for input-level alignment, while only a few measured target-tool S_a labels are used for output-level affine calibration. This design addresses the practical metrology constraint that target-tool process signals are easily acquired, whereas dense target-tool surface topography measurements are costly. Therefore, the contribution of this work lies in the measurement-driven problem formulation, pass-level data organization, low-label target-tool calibration protocol, and systematic comparison of descriptor-based and fusion-based predictors under identical LOTO settings.

The main contributions of this work are summarized as follows:

1. A measurement-driven pass-level S_a prediction framework is established for Ti–6Al–4V milling after tool replacement. Each steady-state cutting pass is linked to a single measured S_a value, providing a direct correspondence between process signals and surface-topography-derived quality labels.
2. A one-pass–one-sample data organization strategy is used to avoid overlapping window sample inflation. The proposed organization explicitly distinguishes pass-level S_a labels from layer-level tool-life progression and provides a reproducible basis for evaluating prediction reliability under a tool-to-tool shift.
3. A measurement-driven low-label target-tool adaptation setting is formulated for S_a prediction after tool replacement. Unlike generic source-only transfer or conventional domain-adaptation settings, the target-tool unlabeled descriptors are used only for input-level normalization, while a few measured target-tool S_a values are used only for affine output calibration. This design reflects practical machining conditions in which full target-tool roughness measurement is costly, but limited metrology feedback can be obtained.
4. Statistical-descriptor-based regressors and a hierarchical signal statistical fusion comparison model are evaluated under identical LOTO partitions, normalization strategies, and calibration protocols. The comparison clarifies whether compact descriptors and conventional regressors are more suitable than a more complex fusion architecture under the present controlled three-tool small-sample setting.
5. Pass-uniform, layer-uniform all-pass, and layer-uniform one-pass calibration protocols are systematically compared to clarify differences in label availability across target-tool scenarios. These protocols distinguish a controlled full-life calibration upper bound from more conservative layer-wise adaptation settings, helping avoid overly optimistic conclusions in low-label target-tool evaluation.

2 Experiments and Data

2.1 Experimental Setup and Signal Acquisition

End-milling experiments on TC4 (Ti–6Al–4V) were conducted using a five-axis CNC machining center (Jingdiao JDGR400T). The workpiece size was 50 mm × 50 mm × 30 mm, and a four-flute shank radius end mill (RM-4R-D10.0R1.0) was used. The main tool specifications and cutting parameters are summarized in Table 1.

During milling, cutting force and vibration signals were synchronously acquired. The cutting forces were measured using a triaxial dynamometer (ME-Systeme K3D160), providing the cutting force components F_x , F_y , and F_z . The vibration signal Vib was collected using an IEPE single-axis piezoelectric accelerometer [26, 27]. After signal conditioning, all channels were recorded by an MI-7016 data acquisition system at a sampling frequency of 5 kHz [28, 29]. The dynamometer was mounted between the fixture and the machine table, while the accelerometer was attached to the fixture to capture the dynamic response during cutting.

The experimental platform and signal acquisition system are shown in Fig. 1.

2.2 Surface Roughness and Tool-Wear Characterization

Surface topography was measured using a laser confocal white-light interferometer (Sensofar S neox), and S_a was adopted as the pass-level surface-quality label [30, 31]. For each machined pass, the measurement region was selected from the middle stable cutting area of the corresponding milled surface, avoiding the entry edge, exit edge, and boundary region. A stitched surface topography region of approximately 12.34 mm × 5.21 mm was acquired using a

50× objective. Before the areal roughness evaluation, the acquired topography was processed by tilt correction in the SensoVIEW software. The parameter S_a was then evaluated following ISO 25178. No additional task-specific S-filter or L-filter cutoff was manually specified beyond the standard evaluation settings used in the instrument software. For each pass, three repeated surface topography measurements were performed, taking their average value as the final pass-level S_a label. The average S_a value was then aligned with the corresponding pass-level steady-state force/vibration signal sample. Although repeated measurements were used to reduce random measurement variation, the repeated-measurement uncertainty was not separately quantified as an independent uncertainty component in

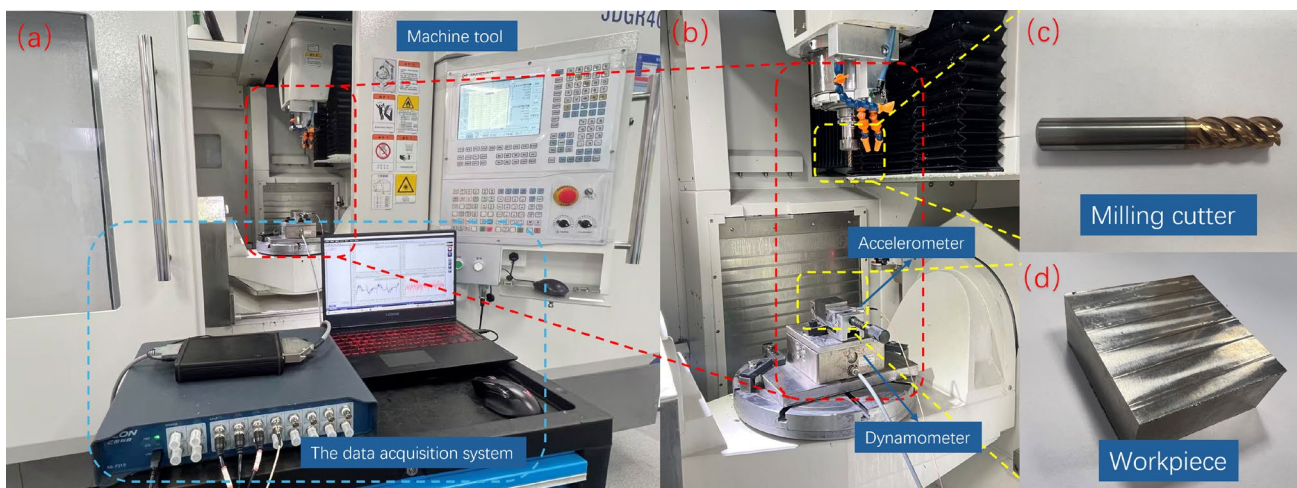


Fig. 1 Experimental setup and measurement system: **a** milling experimental platform (machine tool/tool/fixture/dynamometer); **b** accelerometer and vibration sensor; **c** four-flute end mill; and **d** Ti-6Al-4V workpiece (50 mm × 50 mm × 30 mm)

Table 1 Tool specifications and cutting parameters used in the TC4 milling experiments

Category	Parameter	Value
Tool type	Four-flute shank radius end mill	RM-4R-D10.0R1.0
Coating	Coating material	TiAlXN
Geometry	Tool diameter D	10 mm
Geometry	Corner radius R	1.0 mm
Geometry	Shank diameter d	10 mm
Geometry	Cutting edge length H	25 mm
Geometry	Overall length L	75 mm
Geometry	Helix angle β	42°
Structure	Number of flutes z	4
Cutting parameter	Spindle speed n	1600 r/min
Cutting parameter	Feed per tooth f_z	0.07 mm/tooth
Cutting parameter	Feed speed v_f	448 mm/min
Cutting parameter	Cutting speed v_c	50.27 m/min
Cutting parameter	Axial depth of cut a_p	1.0 mm
Cutting parameter	Radial depth of cut a_e	9.0 mm
Process condition	Machining process / cooling	End milling / dry cutting

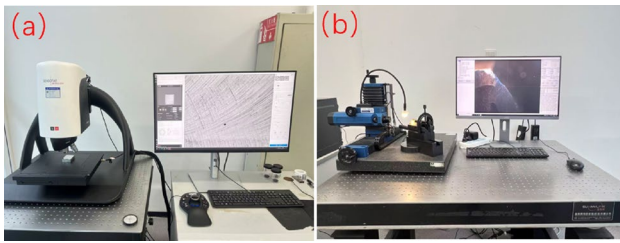


Fig. 2 Measurement devices: **a** laser confocal white-light interferometer for surface topography measurement and **b** tool measuring instrument for flank wear evaluation

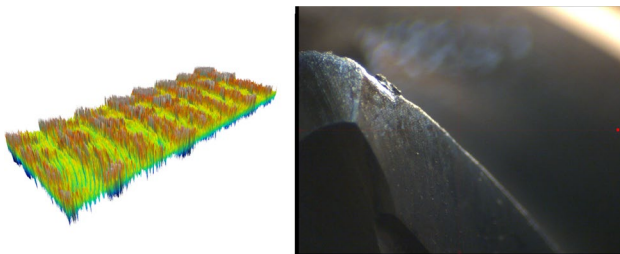


Fig. 3 Examples of measurement results: **a** 3D surface topography with extracted S_a and **b** a flank wear measurement interface for VB evaluation

the present study. The surface measurement device and an example of the extracted S_a result are shown in Figs. 2 and 3.

To characterize tool wear, the flank wear land width VB was measured using a tool measuring instrument (Omega PG1000–400–4K). Because non-uniform wear may occur on multi-flute tools, VB was measured for all four cutting edges and recorded at the layer level [32]. Since wear was measured once for each machined layer, the corresponding VB value was shared by the valid passes within each layer. In this study, tool replacement and progressive tool wear are treated as related but different sources of machining state variation. Tool replacement mainly refers to the distribution shift between different tools, whereas progressive tool wear describes the within-tool cutting state evolution over the tool's lifetime. Therefore, the measured VB values were used to characterize tool-life progression, verify that the dataset spanned a broad wear range, and support the physical interpretation of changes in surface morphology. However, VB was not used as a supervised input or auxiliary prediction target in the final selected prediction model. This choice was made because S_a was measured at the pass level, whereas VB was measured at the layer level. Directly using layer-level VB as a pass-level model input would repeatedly assign identical wear values to multiple passes within a layer and could introduce an artificial form of repeated layer-level supervision. Therefore, the modeling task examines

pass-level S_a prediction using signal statistical descriptors and low-label target-tool adaptation.

Representative examples of tool wear and corresponding surface topography are shown in Fig. 4. As the tool wear progresses from slight to more severe, VB becomes more evident, and the surface morphology tends to deteriorate. This observation supports the physical relevance of tool-life progression to surface generation. However, this figure is used only to characterize the experimental dataset and provide machining context interpretation; the final selected prediction model does not use VB or wear stage labels as supervised inputs.

2.3 Signal Preprocessing and Steady-State Sample Construction

A unified preprocessing pipeline was applied to the recorded F_x , F_y , F_z , and V_{ib} signals before sample construction. The purpose was to reduce sensor drift, non-cutting transients, and isolated spike-like artifacts while preserving the stable cutting dynamics used for roughness prediction.

First, baseline removal was performed for each channel to suppress DC offset:

$$x_d(t) = x(t) - \bar{x}, \quad (1)$$

where $x_d(t)$ denotes the baseline-corrected signal after DC-offset removal, $x(t)$ denotes the original signal, and $\bar{x}$ is its mean value. This operation was applied consistently to all force and vibration channels before steady-state extraction [33, 34].

Second, steady-state cutting intervals were extracted automatically. A sliding envelope based on the amplitude or moving root-mean-square (RMS) of F_x was first used to locate the coarse cutting intervals corresponding to the six tool passes in each layer. Within each coarse interval, a steady-state gating procedure based on high-frequency energy indicators was further used to retain the stable cutting plateau and remove entry, exit, and residual transient portions. The first 5% and last 5% of the retained plateau were then trimmed to reduce boundary effects. A comparison between the raw multichannel signals and the extracted steady-state segments is shown in Fig. 5.

Third, isolated spike-like artifacts were corrected conservatively. For each valid pass and each channel, a Hampel filter was applied independently using a sliding window. The median absolute deviation (MAD) and robust scale estimate are defined as

$$\begin{aligned} \text{MAD} &= \text{median}(|x_i - \text{median}(x_i)|), \\ \sigma_{\text{rob}} &= 1.4826 \text{MAD}. \end{aligned} \quad (2)$$

A sample x_i was identified as an outlier when

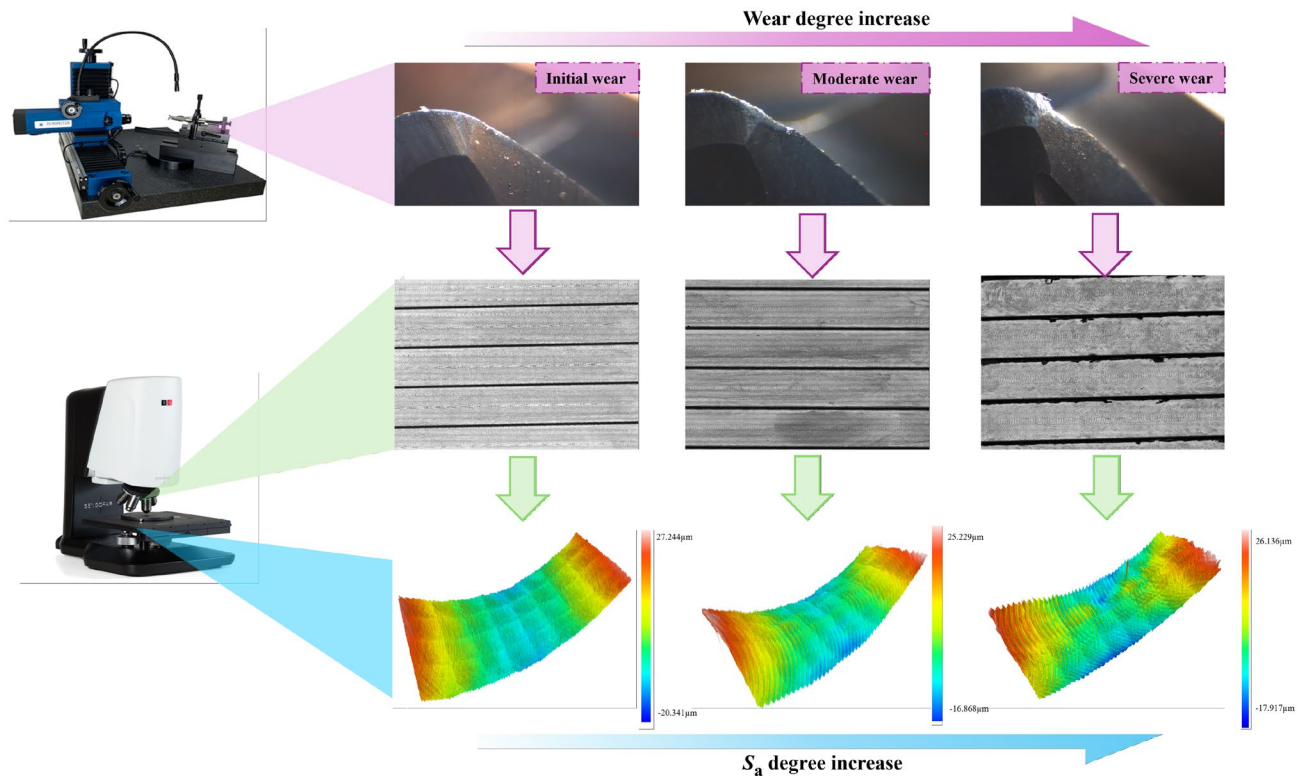


Fig. 4 Representative correspondence between tool wear states and surface topography characteristics. The examples illustrate that tool-life progression is associated with changes in VB and machined sur-

face morphology. In this study, VB is used for dataset characterization and mechanism interpretation, rather than as a supervised input to the final selected prediction model

$$|x_i - \text{med}_w| > \eta \sigma_{\text{rob}}, \quad (3)$$

where med_w denotes the local median calculated within a sliding window of length w , w is the Hampel-filter window length in samples, and η is the outlier threshold coefficient. In this study, conservative parameters of $w = 41$ samples and $\eta = 4.0$ were used. Only the detected outlier points were repaired by linear interpolation within the same pass; boundary points were filled using the nearest valid value. No global smoothing was applied, so normal tooth-passing fluctuation and cutting impacts were not suppressed [35].

2.4 Pass-Level Sample Definition and Dataset Statistics

Each machining layer contains six tool passes. After pre-processing and steady-state extraction, each pass is treated as one sample source. This one-pass-one-sample strategy establishes a direct correspondence between the cutting signal and the machined surface unit and avoids artificial sample inflation caused by overlapping sliding-window segmentation [36, 37].

Because the effective length N of each steady-state segment may vary, a fixed-length rule was used to construct

uniform model inputs. If $N \geq L$, the middle $L = 384$ points of each channel were retained by center cropping:

$$s = \left\lfloor \frac{N-L}{2} \right\rfloor, \quad x_c^{\text{crop}} = x_c[s : s + L - 1], \quad (4)$$

where s denotes the starting index of the cropped sequence, x_c denotes the one-dimensional signal of channel c , and x_c^{crop} denotes the cropped sequence of that channel. If $N < 384$, the pass was discarded. No padding, interpolation-based extension, or resampling was used for short sequences, so as to avoid introducing artificial temporal information. The cropped sequences from all four synchronized channels are then stacked to form the waveform sample:

$$X^{(i,p)} \in \mathbb{R}^{C \times L}, \quad C = 4, \quad L = 384. \quad (5)$$

where, i and p denote the layer index and pass index, respectively; C denotes the number of signal channels ($C=4$); and L denotes the temporal sequence length ($L=384$). A binary field, `keep_flag`, was used in the master label table to indicate whether a pass sample satisfied the fixed-length requirement.

Areal surface roughness was characterized by S_a (μm), measured using a laser confocal white-light interferometer.

Steady cutting segment extraction results (pass 1–6)

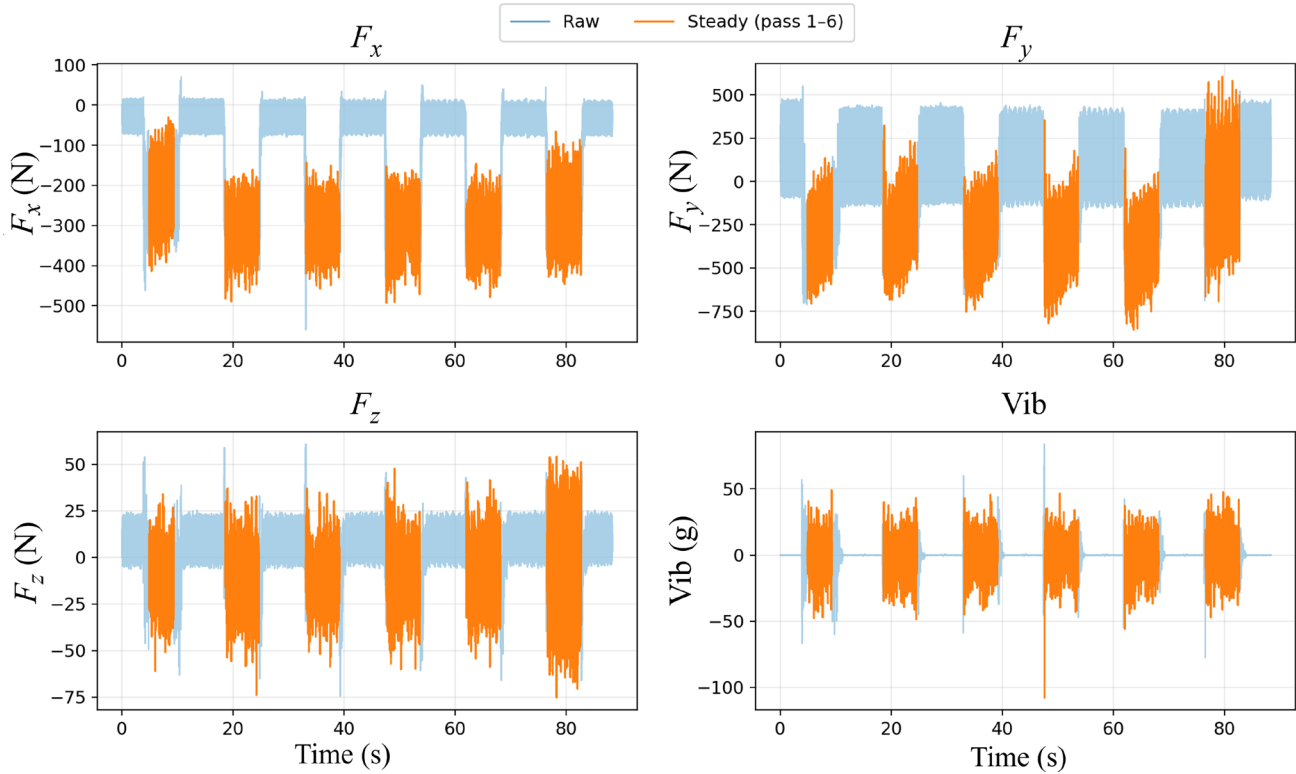


Fig. 5 Comparison between raw multi-channel cutting signals and automatically extracted steady-state cutting segments for passes 1–6. The raw signals include entry, exit, and non-cutting intervals, whereas the extracted steady-state segments correspond to the stable

cutting portions retained for pass-level sample construction. This figure demonstrates the signal portions used to construct one-pass-one-sample inputs for subsequent descriptor extraction and model evaluation

For each machined pass, three repeated surface topography measurements were conducted in the middle stable cutting region, and the average S_a value was used as the final pass-level roughness label. One average S_a value was assigned to each machined pass, giving a one-to-one correspondence between each valid signal sample and its roughness label:

$$y_{S_a}^{(i,p)} = S_a^{(i,p)}, \quad (6)$$

The label was uniquely indexed by tool ID, layer ID, pass ID, and file name.

Tool wear was measured at the layer level using VB. Since VB was measured once per layer, the same layer-level wear value was applied to all valid passes in each layer. Accordingly, VB was treated as a layer-level descriptor for dataset characterization rather than as a pass-level supervised model input. This distinction prevents layer-level wear measurements from being repeated as if they were independent pass-level labels. Thus, each valid sample can be represented as

$$\left(\mathbf{X}^{(i,p)}, y_{S_a}^{(i,p)}, \text{VB}^{(i)} \right), \quad (7)$$

where $\text{VB}^{(i)}$ denotes the layer-level wear descriptor used only for dataset characterization and mechanism interpretation, not as a supervised input for the final S_a prediction.

After the fixed-length rule was applied, the three tools contained 718 valid pass-level samples out of 744 original pass records. Specifically, C1, C2, and C3 contained 251, 240, and 227 valid samples, respectively. Twenty-six samples were removed because their steady-state lengths were shorter than 384 points, corresponding to an overall

Table 2 Dataset statistics after fixed-length pass-level sample construction

Tool	Layers	Pass records	Valid samples	Invalid	Removal rate (%)
C1	44	264	251	13	4.92
C2	42	252	240	12	4.76
C3	38	228	227	1	0.44
Total	124	744	718	26	3.49

removal rate of 3.49%. The dataset statistics are summarized in Table 2.

The roughness and wear distributions of the retained samples are shown in Fig. 6. The median S_a is 1.053 μm , with an interquartile range of 0.298 μm . The median VB is 0.168, with an interquartile range of 0.112 and a range of 0.020–0.544. Because VB was measured at the layer level and shared by valid passes within a layer, its distribution is counted at the pass-sample level only for consistency with the modeling samples. These statistics indicate that the dataset spans a broad tool-life range and provides a controlled basis for evaluating tool-to-tool shifts under fixed cutting parameters.

3 Method

3.1 Task Definition and Low-Label Adaptation Pipeline

The objective of this study is to predict pass-level S_a from force/vibration signals under a tool-replacement-induced distribution shift. Since S_a is obtained from offline surface topography measurement, the prediction model is expected to provide a signal-based estimate of the measured surface quality value rather than replace the metrology process itself [30, 31]. In practical machining, a model trained on source tools may perform poorly when applied directly to a newly introduced target tool, even when the cutting parameters remain unchanged. This degradation is mainly caused by tool-to-tool differences in signal scale, spectral response, and output S_a distribution [20]. Therefore, this work does not formulate the task as a strict source-only

generalization. Instead, it examines a low-label target-tool adaptation setting for measurement-driven S_a prediction after tool replacement [21].

In each LOTO fold, two tools serve as source tools, and the remaining tool serves as the unseen target tool. The source tools provide labeled pass-level samples for model training. The target tool provides unlabeled signal/statistical-feature distributions for input-level normalization, and only a few labeled S_a samples are used for affine output calibration. No target-tool S_a labels are used during source model training, descriptor ranking, or per-tool feature normalization.

The overall pipeline is summarized in Fig. 7 and contains five sequential stages. First, experimental measurements provide synchronized force/vibration signals and pass-level measured S_a labels. Second, steady-state pass-level samples are extracted from the multi-channel signals using a one-pass-one-sample strategy. Third, each valid pass is represented by all 72 statistical descriptors, layer-level mean descriptors, and pass-index information. Fourth, descriptor-based regressors and the hierarchical signal statistical fusion comparison model are trained on source tools under the LOTO setting, while unlabeled target-tool descriptors are used for per-tool input normalization. Fifth, few-shot affine calibration is applied using limited target-tool S_a label measurements, and the calibrated model is evaluated on the remaining target-tool samples.

3.2 Pass-Level Data Representation

Each machining layer contains up to six cutting passes. After baseline removal, steady-state segment extraction, and outlier correction, each valid pass is converted into a fixed-length four-channel signal sequence. The four channels

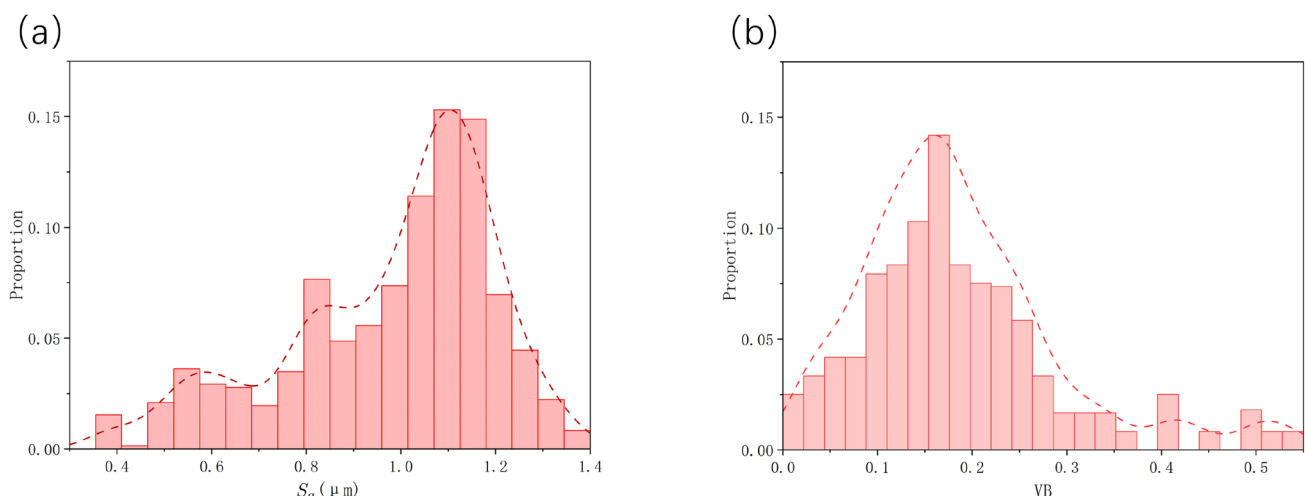


Fig. 6 Distributions of retained pass-level samples from C1, C2, and C3: **a** areal surface roughness S_a and **b** layer-level VB counted at the pass-sample level for consistency with the modeling samples

comprise F_x , F_y , F_z , and Vib. The waveform input of the p -th pass in the i -th layer is denoted as

$$X_{i,p} \in \mathbb{R}^{C \times L}, \quad C = 4, \quad L = 384. \quad (8)$$

When the steady-state segment exceeds 384 points, center cropping is used. Samples shorter than 384 points are discarded rather than padded or resampled.

The corresponding pass-level roughness label is

$$y_{i,p} = S_{a,i,p}, \quad (9)$$

where $S_{a,i,p}$ is the measured S_a of the corresponding machined pass. This one-pass-one-sample definition avoids the artificial sample inflation caused by overlapping sliding windows and provides a clear correspondence between signal input and roughness label [38].

Because some layers contain fewer than six valid passes after fixed-length screening, a binary validity mask is introduced:

$$m_{i,p} = \begin{cases} 1, & \text{if pass } p \text{ of layer } i \text{ is valid,} \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Only valid passes with $m_{i,p} = 1$ are used for training, calibration, and evaluation. The mask is also used to compute layer-level mean descriptors and masked layer aggregation so that invalid passes are not treated as real samples.

Tool wear, represented by VB, is measured at the layer level and retained for tool-life characterization and mechanism discussion. However, VB and wear stage labels are not used as supervised inputs or auxiliary prediction targets in the final selected prediction model. This design separates two roles of wear information: VB is used to describe the physical tool-life state of the experimental dataset, whereas the prediction model estimates pass-level S_a from signals and statistical descriptors that are available for each pass. Because VB is not measured independently for each pass, using it as a pass-level input imposes repeated layer-level information on multiple pass-level samples. This approach avoids imposing layer-level wear information as repeated pass-level supervision and keeps the prediction task focused on pass-level S_a .

3.3 Pass-Layer Organization and Layer Context Descriptors

The dataset has a natural pass-layer structure. A tool contains multiple machined layers, each layer with several pass-level samples. Although S_a is measured at the pass level, the passes within a layer are not completely independent because they share similar tool-life states, machining environments, and layer-level wear conditions. Therefore, the pass-layer organization is explicitly retained in the data representation.

For the i -th layer, the set of pass-level waveform samples and the corresponding mask vector are expressed as

$$X_i = \{X_{i,p}\}_{p=1}^6, \quad (11)$$

and

$$m_i = [m_{i,1}, m_{i,2}, \dots, m_{i,6}]. \quad (12)$$

This structure is used in two ways. For descriptor-based regressors, layer-level mean descriptors are calculated from the valid passes within a layer and concatenated with the pass-level descriptor vector. For the fusion comparison model, the same validity mask is used in the masked layer aggregation module. Thus, the layer context is introduced without generating additional overlapping samples or treating invalid passes as real observations.

This representation is especially important for small-sample cross-tool evaluation. It preserves the real pass-layer data organization, avoids excessive sample inflation, and provides a consistent input structure for statistical descriptor regressors and the hierarchical fusion comparison model [39].

3.4 Pass-Level Statistical Descriptors

For each valid steady-state pass, statistical descriptors are extracted from the four-channel force/vibration signals to provide a compact description of the cutting state. These descriptors characterize amplitude, energy, fluctuation intensity, distribution shape, and spectral response. Typical descriptors include the mean value, standard deviation, RMS, mean absolute value, peak-to-peak value, extrema, spectral centroid, and spectral-energy-related quantities. The descriptor vector of pass p in layer i is denoted as

$$s_{i,p} \in \mathbb{R}^D, \quad (13)$$

where D is the total number of candidate descriptors extracted from the valid steady-state signal.

To avoid target-label leakage, descriptor ranking is performed only on the source-domain training data within each LOTO fold. For the d -th descriptor, the absolute Pearson correlation between the descriptor values and the training-set S_a labels is calculated as

$$r_d = \left| \text{corr}(s_d, y_{S_a}) \right|, \quad d = 1, 2, \dots, D. \quad (14)$$

where r_d is the absolute correlation coefficient of the d -th descriptor, s_d denotes the vector of the d -th descriptor values over all source-training samples, y_{S_a} denotes the corresponding vector of training-set S_a labels, and D is the total number of candidate descriptors. The descriptors are then ranked in descending order of r_d :

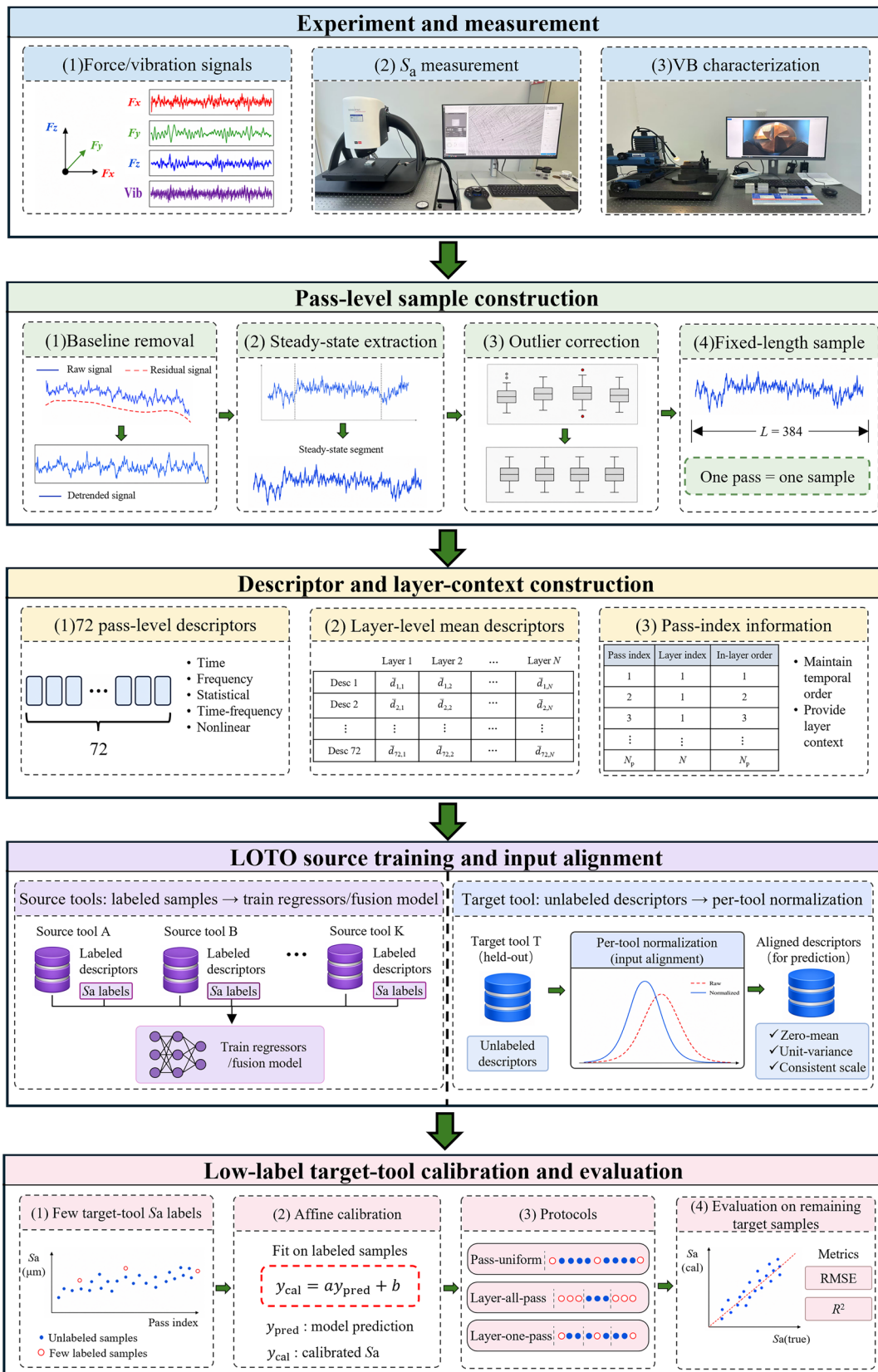


Fig. 7 Overall workflow of the low-label target-tool adaptation framework. Pass-level steady-state samples are constructed from multi-channel force/vibration signals. All 72 statistical descriptors, layer-level mean descriptors, and pass index information are used to construct descriptor-based regressors. Additionally, a hierarchical signal statistical fusion model is retained as a structured comparison model. In each LOTO fold, labeled source-tool samples are used for model training, unlabeled target-tool descriptors are used for per-tool input normalization, limited measured target-tool S_a labels are used only for affine output calibration, and the remaining target-tool samples are used for final evaluation. The trained source models are evaluated using raw LOTO prediction and fixed-budget affine calibration protocols

$$\mathcal{R} = \text{argsort}(r_d, \text{descending}). \quad (15)$$

This ranking is used only for fold-wise descriptor analysis and for assessing the influence of descriptor retention. The final descriptor-based regressors retain all available descriptors. Since the candidate pool contains 72 descriptors, the final pass-level descriptor vector is written as

$$\tilde{s}_{i,p} \in \mathbb{R}^{72}. \quad (16)$$

This all-descriptor setting is adopted to avoid unstable fold-dependent feature selection under the small three-tool LOTO setting. Since the available source tools change across folds, selecting a fixed top- K subset may make the retained descriptors sensitive to the specific source-tool combination. Tree-based ensemble models such as RF and LSBoost reduce the influence of redundant or weakly relevant descriptors through their split-selection and ensemble mechanisms. Therefore, all 72 descriptors are retained to provide a stable and reproducible representation for model comparison. Descriptor ranking is used only for fold-wise analysis and does not involve target-tool S_a labels [40].

3.5 Per-Tool Unsupervised Feature Normalization

Tool replacement can change the scale and distribution of statistical descriptors, even when the cutting parameters remain unchanged. To reduce this input-level mismatch, per-tool unsupervised normalization is applied to the 72 statistical descriptors. For the source tools, the normalization statistics are calculated using the valid source-tool passes available in the corresponding LOTO fold. In the present study, the target tool's normalization statistics are calculated using all valid unlabeled passes collected over its life. Therefore, this step should be interpreted as an offline unsupervised input-level alignment strategy, rather than an early-stage online normalization strategy. For each tool q , the mean and standard deviation of the d -th descriptor are calculated as follows:

$$\mu_{q,d} = \frac{1}{N_q} \sum_{(i,p) \in \Omega_q} \tilde{s}_{i,p,d}, \quad (17)$$

$$\sigma_{q,d} = \sqrt{\frac{1}{N_q - 1} \sum_{(i,p) \in \Omega_q} (\tilde{s}_{i,p,d} - \mu_{q,d})^2}, \quad (18)$$

where, $\tilde{s}_{i,p,d}$ denotes the d -th component of the 72-dimensional pass-level statistical descriptor vector $\tilde{s}_{i,p}$ for pass p in layer i , with $d = 1, 2, \dots, 72$; Ω_q denotes the set of valid pass samples belonging to q and N_q is the number of valid passes in this tool. The normalized descriptor is obtained as

$$\bar{s}_{i,p,d} = \frac{\tilde{s}_{i,p,d} - \mu_{q,d}}{\sigma_{q,d} + \epsilon}, \quad (19)$$

where ϵ is a small constant used to avoid division by zero. The normalized values are clipped to suppress extreme values before model training and evaluation.

This normalization uses only unlabeled signal/statistical feature distributions and does not use target-tool S_a labels. Therefore, it should not be interpreted as label leakage. However, because the target-tool normalization statistics are estimated from the full available unlabeled target-tool descriptor distribution, the raw setting in this study is not a strict source-only or early-stage online setting. It is more accurately described as cross-tool prediction with offline unsupervised input-level alignment, followed by limited-label affine output calibration. In practical online deployment, the same idea would require rolling or incremental estimation of target-tool normalization statistics from newly acquired unlabeled signals. This distinction is important for practical target-tool adaptation and for avoiding overly optimistic claims about source-only transfer [41].

The role of this normalization step is limited to input-level alignment. Tool replacement may change the magnitude range, variance, and distribution shape of force/vibration-derived descriptors even under fixed cutting parameters. Consequently, per-tool normalization reduces descriptor-scale mismatch between source tools and the target tool using only unlabeled target-tool information. However, this operation does not use measured target-tool S_a labels and cannot correct the target-tool output response itself. In other words, per-tool normalization can reduce input distribution shift, but it cannot remove systematic prediction offset or scale error in the predicted S_a . Thus, an additional output-level calibration step is required.

3.6 Hierarchical Signal Statistical Fusion Model for Comparison

In addition to descriptor-based regressors, a hierarchical signal statistical fusion model is constructed as a structured

deep learning comparison architecture. This model examines whether waveform representation and masked layer aggregation provide additional information compared with statistical descriptors under the same LOTO, normalization, and calibration settings. Therefore, it is used for fair comparison rather than being predefined as the final proposed model.

The fusion model comprises a CNN waveform branch, a statistical feature branch, a masked layer aggregation module, and a pass-level regression head. Its architecture is shown in Fig. 8.

For each valid pass, the fixed-length waveform input $X_{i,p} \in \mathbb{R}^{4 \times 384}$ is first standardized by sample-wise and channel-wise normalization:

$$X'_{i,p,k} = \frac{X_{i,p,k} - \text{mean}(X_{i,p,k})}{\text{std}(X_{i,p,k}) + \epsilon}, \quad (20)$$

where, $X'_{i,p,k}$ denotes the normalized waveform of the k -th signal channel for pass p in layer i ; $X_{i,p,k}$ denotes the corresponding original waveform; $\text{mean}(X_{i,p,k})$ and $\text{std}(X_{i,p,k})$ denote the mean and standard deviation calculated over the temporal samples of that channel, respectively; k denotes the signal-channel index, $k = 1, 2, 3, 4$. The normalized waveform is fed into a lightweight CNN encoder:

$$h_{i,p}^{\text{cnn}} = f_{\text{cnn}}(X'_{i,p}; \theta_{\text{cnn}}), \quad (21)$$

where $h_{i,p}^{\text{cnn}}$ is the waveform embedding; θ_{cnn} denotes the trainable parameters of the CNN encoder; $X'_{i,p}$ denotes the normalized four-channel waveform matrix of pass p in layer i , obtained using the normalization defined in Eq. (20); $f_{\text{cnn}}(\cdot)$ denotes the nonlinear mapping implemented by the lightweight CNN encoder.

The per-tool-normalized statistical descriptor vector $\bar{s}_{i,p} \in \mathbb{R}^{72}$ is encoded by a fully connected statistical branch:

$$h_{i,p}^{\text{stat}} = f_{\text{stat}}(\bar{s}_{i,p}; \theta_{\text{stat}}), \quad (22)$$

where $h_{i,p}^{\text{stat}}$ is the statistical-feature embedding, θ_{stat} denotes the trainable parameters of the statistical branch, $f_{\text{stat}}(\cdot)$ denotes the nonlinear mapping implemented by the fully connected statistical branch. The waveform embedding and statistical feature embedding are concatenated to form the pass-level representation:

$$h_{i,p} = [h_{i,p}^{\text{cnn}}; h_{i,p}^{\text{stat}}]. \quad (23)$$

To preserve the natural pass-layer structure and avoid invalid-pass interference, masked layer aggregation is performed over valid passes within a layer:

$$z_i = \frac{\sum_{p=1}^6 m_{i,p} h_{i,p}}{\sum_{p=1}^6 m_{i,p} + \epsilon}, \quad (24)$$

where z_i denotes the layer-level context embedding, $m_{i,p}$ is the binary validity mask of pass p in layer i , $p = 1, 2, \dots, 6$ denotes the pass index within the layer. The final regression input comprises the pass embedding, layer context, and pass-index one-hot vector:

$$q_{i,p} = [h_{i,p}; z_i; e_p], \quad (25)$$

where e_p represents the one-hot encoding of the pass index. The predicted roughness is then obtained by

$$\hat{y}_{i,p} = f_{\text{reg}}(q_{i,p}; \theta_{\text{reg}}). \quad (26)$$

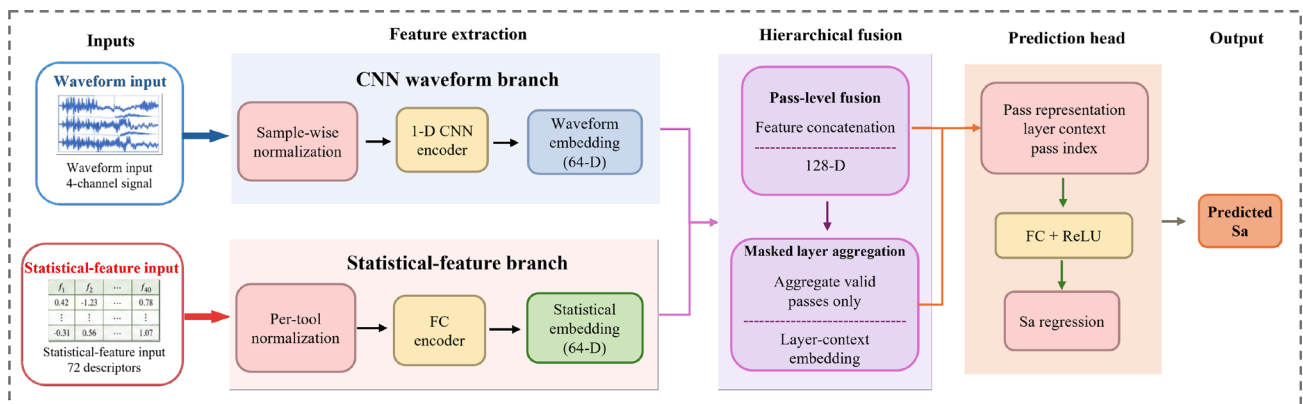


Fig. 8 Architecture of the hierarchical signal statistical fusion model used as a structured deep learning comparison model. The model combines a CNN waveform branch for the fixed-length four-channel steady-state signal and a statistical feature branch for all 72 pass-level descriptors. The pass-level embeddings are fused and further aggre-

gated by masked layer pooling to generate layer context information exclusively from valid passes. The pass-index one-hot vector is then combined with the pass-level and layer context embeddings for the final S_a regression

where θ_{reg} denotes the trainable parameters of the regression head. The fusion model is trained by minimizing the mean squared error over valid pass samples:

$$\mathcal{L}_{S_a} = \frac{\sum_i \sum_{p=1}^6 m_{i,p} (\hat{y}_{i,p} - y_{i,p})^2}{\sum_i \sum_{p=1}^6 m_{i,p}}. \quad (27)$$

Tool wear is not used as an auxiliary supervised target in the final comparison. This approach avoids imposing layer-level wear labels on pass-level S_a prediction, keeping the comparison focused on signal statistical representation and low-label target-tool adaptation.

3.7 Training and Validation of the Fusion Comparison Model

The fusion comparison model is trained only on source-tool samples in each LOTO fold. Between the two source tools, approximately 20% of the layers are uniformly selected from the tool's lifespan as the validation set, and the remaining layers are used for training. This layer-wise validation split reduces the risk that validation samples are concentrated within a narrow tool-life interval.

The training set S_a labels are standardized by z -score normalization:

$$y' = \frac{y - \mu_y}{\sigma_y}, \quad (28)$$

where, y denotes the original measured S_a label in the source training set, y' denotes the corresponding standardized label, μ_y and σ_y denote the mean and standard deviation of the source-training S_a labels, respectively. Predictions are transformed back to the original S_a scale during evaluation. Balanced resampling based on roughness bins is applied during training to reduce label distribution imbalance. The Adam optimizer is used for parameter optimization, and early stopping is applied according to the source validation root mean squared error (RMSE).

Because the dataset contains only three tools and the pass-level samples are limited, random initialization may affect the fusion model. Therefore, five independent fusion models are trained using different random seeds. Let $\hat{y}_{i,p}^{(m)}$ denote the prediction of the m -th seed model, where $m = 1, 2, \dots, M$ and $M = 5$. The final fusion model prediction is obtained by mean-of-all-seeds averaging:

$$\hat{y}_{i,p}^{\text{ens}} = \frac{1}{M} \sum_{m=1}^M \hat{y}_{i,p}^{(m)}. \quad (29)$$

This ensemble strategy does not use target-tool test labels or calibration labels for seed selection. The strategy is used only to reduce the training variance of the fusion comparison

model. Similar ensemble ideas have been adopted in data-driven machining monitoring and industrial prediction tasks to improve robustness [42, 43].

3.8 Descriptor-Based Regressors

Descriptor-based regressors are constructed to evaluate whether compact statistical descriptors can provide reliable cross-tool S_a prediction under limited target-tool calibration. For each valid pass, the input vector comprises three parts: the 72 per-tool-normalized pass-level statistical descriptors, the corresponding layer-level mean descriptors, and the pass-index one-hot vector. The layer-level mean descriptors summarize the valid passes within a layer, while the pass-index vector preserves the toolpath-order information.

The descriptor-regressor input can be expressed as

$$\mathbf{x}_{i,p}^{\text{ML}} = [\bar{\mathbf{s}}_{i,p}; \bar{\mathbf{s}}_i^{\text{layer}}; \mathbf{e}_p], \quad (30)$$

where $\bar{\mathbf{s}}_i^{\text{layer}}$ denotes the mean descriptor vector of valid passes in layer i . The vectors $\bar{\mathbf{s}}_{i,p}$ and $\mathbf{e}_p$ have been defined above as the normalized pass-level descriptor vector and the pass-index one-hot vector, respectively. Therefore, the descriptor-regressor input includes pass-level information, layer-context information, and pass identity.

The compared descriptor-based regressors include support vector regression with a radial basis function kernel (SVR-RBF), support vector regression with a linear kernel (SVR-Linear), RF, GPR, and LSBoost. All descriptor-based regressors are evaluated under the same LOTO partition, per-tool unsupervised feature normalization, and affine-calibration protocols as the fusion comparison model. This design ensures that the comparison reflects differences in model structure rather than differences in data partitioning, normalization, or calibration strategy [42, 43].

3.9 Few-Shot Affine Calibration

Raw cross-tool prediction may still contain systematic output bias and scale mismatch after per-tool unsupervised normalization. This outcome arises because input-level descriptor alignment only reduces differences in the statistical feature distribution, whereas the relationship between the aligned descriptors and the measured S_a response can still vary across tools. In the present measurement-driven setting, collecting a dense set of target-tool surface topography measurements is costly. However, a few measured target-tool S_a values can be obtained for calibration. Therefore, affine calibration is used as a simple and interpretable output-level correction strategy for low-label target-tool adaptation. The affine form corrects the dominant offset and scale errors in the transferred prediction without introducing a complex target-domain model that may overfit the limited calibration

samples. Given a small labeled calibration set from the target tool, an affine mapping is fitted between the raw prediction $\hat{y}$ and the measured roughness y :

$$\hat{y}^{\text{cal}} = a\hat{y} + b, \quad (31)$$

where a and b are estimated by least-squares fitting on the target-tool calibration set, $\hat{y}$ denotes the raw model prediction, and $\hat{y}^{\text{cal}}$ denotes the calibrated prediction.

$$(a, b) = \arg \min_{a, b} \sum_{j \in \mathcal{C}} (y_j - a\hat{y}_j - b)^2, \quad (32)$$

where $\mathcal{C}$ denotes the calibration set. After the affine parameters are fitted, the calibration samples are excluded from the final evaluation set, and the fitted mapping is applied only to the remaining target-tool evaluation samples. This rule is used consistently for all calibration protocols to avoid evaluating the model on samples used for output correction.

Three fixed-budget calibration protocols are used in this study: pass-uniform calibration with $N_{\text{pass}} = 30$, layer-uniform all-pass calibration with $N_{\text{layer}} = 10$, and layer-uniform one-pass calibration with $N_{\text{layer}} = 10$. These protocols are designed to represent different levels of target-tool measurement availability. The pass-uniform protocol selects calibration passes uniformly throughout the available target-tool life. Therefore, it represents a controlled full-life adaptation upper-bound case, rather than a strictly online early-stage deployment case. The layer-uniform all-pass protocol assumes that several representative target-tool layers are inspected and that all valid passes in these layers can be measured. The layer-uniform one-pass protocol is more conservative because only one valid pass from each selected layer is used for calibration, which reduces the influence of within-layer pass correlation and better reflects a sparse low-label calibration condition. The detailed evaluation setting is described in Sect. 4.1. Similar low-label calibration strategies have been used to correct target-domain response shifts in sensor-based prediction tasks [44].

3.10 Evaluation Metrics

The prediction performance is evaluated using RMSE and the coefficient of determination R^2 :

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2}, \quad (33)$$

$$R^2 = 1 - \frac{\sum_{n=1}^N (y_n - \hat{y}_n)^2}{\sum_{n=1}^N (y_n - \bar{y})^2}. \quad (34)$$

RMSE reflects the absolute prediction error, while R^2 indicates how well the model explains the target-tool roughness variation. In cross-tool prediction, negative R^2 values indicate that using the model is worse than using the mean value of the target-tool samples.

4 Results and Comparative Analysis

4.1 Evaluation Setting

All models are evaluated using a three-fold LOTO protocol. In each fold, one tool is used as the unseen target tool, while the other two tools are used for source-tool training and validation. The final fold assignment is defined as Fold 1 targeting C1, Fold 2 targeting C2, and Fold 3 targeting C3. This setting is designed to examine tool-to-tool shifts under fixed cutting parameters, rather than generalization across different machining conditions. Similar cross-domain and cross-tool distribution shift issues have also been emphasized in recent tool condition monitoring and machining transfer studies [43].

The comparison includes conventional machine learning regressors and the hierarchical signal statistical fusion model. For the conventional regressors, the input features comprise all 72 available pass-level statistical descriptors, the corresponding layer-level mean descriptors, and pass-index one-hot encoding. The compared regressors include SVR-RBF, SVR-Linear, RF, GPR, and LSBoost. For the fusion model, the CNN waveform branch, statistical feature branch, per-tool unsupervised normalization, masked layer aggregation, and mean-of-all-seeds ensemble are used. All models are evaluated using the same LOTO partition, per-tool unsupervised normalization strategy, and affine-calibration protocols.

Because direct raw cross-tool prediction may suffer from systematic output bias and scale mismatch, few-shot affine calibration is further evaluated as a low-label target-tool correction strategy. This evaluation design is used to test whether the proposed affine calibration can correct output-level mismatch under different amounts and organizations of target-tool metrology feedback. Similar few-shot calibration ideas have been used in other sensor-based transfer scenarios to rapidly correct target-domain response shifts with limited labeled samples [44]. Three fixed-budget target-tool calibration protocols are considered:

- **Pass-uniform calibration:** $N_{\text{pass}} = 30$ pass samples are uniformly selected from the target-tool life for affine calibration, and these calibration samples are excluded from the final evaluation set. Because the selected labels are distributed over the whole available target-tool life, this

protocol should be interpreted as a controlled full-life adaptation upper bound. It estimates the potential calibrated performance achievable when limited but representative target-tool labels are available, rather than the performance that can be obtained immediately after a new tool is installed.

- **Layer-uniform all-pass calibration:** $N_{\text{layer}} = 10$ target-tool layers are uniformly selected, and all valid passes in these layers are used for calibration. The selected calibration layers are excluded from the final evaluation set. This protocol corresponds to a stage-wise inspection scenario, in which several representative layers are measured during the target-tool life, and all valid passes in those layers are used to fit the affine correction.
- **Layer-uniform one-pass calibration:** $N_{\text{layer}} = 10$ target-tool layers are uniformly selected, and only one valid pass from each selected layer is used for calibration. The selected calibration layers are excluded from the final evaluation. This protocol is the most conservative among the three settings because it uses fewer pass-level roughness labels and reduces the possible optimistic effect caused by within-layer pass correlation. Therefore, it is used to evaluate sparse low-label target-tool adaptation more strictly.

Notably, these three protocols are not intended to represent the same deployment condition. Instead, they define complementary calibration scenarios with different levels of target-tool label availability: pass-uniform calibration provides an idealized full-life upper-bound estimate, layer-uniform all-pass calibration represents representative stage-wise inspection, and layer-uniform one-pass calibration provides a stricter sparse label evaluation. In all cases, the samples used for affine calibration are excluded from the final evaluation set.

In the following analysis, RF is used for detailed visualization because it achieves the best numerically calibrated performance among the compared models in the present

controlled LOTO setting. However, this selection should not be interpreted as universal RF dominance, because LSBoost and the hierarchical signal statistical fusion model also show competitive calibrated performance. This setting allows the analysis to focus not only on model accuracy but also on the relative roles of statistical descriptors, waveform representation, target-tool calibration, and input-level alignment.

4.2 Comparison with Conventional Machine Learning Baselines

Table 3 summarizes the average LOTO performance of the statistical-descriptor-based regressors and the hierarchical signal statistical fusion model. In the raw cross-tool setting, all methods produce negative R^2 values, indicating that direct signal-based S_a estimation on a newly introduced tool is unreliable even with per-tool unsupervised input normalization. From a measurement adaptation perspective, this result shows that tool replacement changes not only the input descriptor distribution but also the mapping between process signals and the measured surface topography response [20, 41].

After few-shot affine calibration, all models show substantial performance improvements. This finding indicates that a few measured target-tool S_a values can effectively correct the output-level bias of a transferred prediction model [44]. Among the descriptor-based regressors, RF achieves the lowest RMSE under the three calibrated protocols, with RMSE values of 0.0925 μm , 0.0956 μm , and 0.0973 μm under pass-uniform, layer-all-pass, and layer-one-pass calibration, respectively. These results demonstrate that compact statistical descriptors remain highly informative for measurement-driven S_a prediction after tool replacement, especially when they are combined with tree-based ensemble regression and affine output correction. Notably, RF, LSBoost, and the fusion comparison model show modest performance differences under some calibrated protocols. For example, under pass-uniform calibration, RF,

Table 3 Average LOTO performance of statistical-descriptor-based regressors and the hierarchical signal statistical fusion model under raw prediction and fixed-budget affine calibration protocols

Method	Raw RMSE(μm)/ R^2	Pass-uniform RMSE(μm)/ R^2	Layer-all-pass RMSE(μm)/ R^2	Layer-one-pass RMSE(μm)/ R^2
SVR-RBF	0.3026 / -3.9585	0.1106 / 0.2890	0.1085 / 0.2156	0.1107 / 0.1773
SVR-Linear	1.1569 / -110.8032	0.1417 / -0.1161	0.1624 / -0.7997	0.1678 / -0.9335
RF	0.3212 / -4.9426	0.0925 / 0.4658	0.0956 / 0.3331	0.0973 / 0.2908
GPR	0.3282 / -5.0482	0.1014 / 0.3770	0.1071 / 0.1676	0.1075 / 0.1556
LSBoost	0.3167 / -4.8478	0.0952 / 0.4255	0.0988 / 0.2719	0.1063 / 0.1113
Fusion model	0.3074 / -4.2560	0.0949 / 0.4357	0.1008 / 0.2580	0.1046 / 0.1880

Pass-uniform denotes affine calibration using $N_{\text{pass}} = 30$ uniformly selected target-tool pass samples. Layer-all-pass and layer-one-pass use $N_{\text{layer}} = 10$ target-tool layers, with all valid passes and one valid pass from each selected layer used for calibration, respectively. For fair comparison, all results are computed on the same evaluation subset after the calibration samples are excluded. Raw denotes prediction after per-tool unsupervised input-level normalization but before affine output calibration

LSBoost, and the fusion model obtain RMSEs of 0.0925 μm , 0.0952 μm , and 0.0949 μm , respectively. Under layer-all-pass calibration, the corresponding RMSE values are 0.0956 μm for RF, 0.0988 μm for LSBoost, and 0.1008 μm for the fusion model. Therefore, RF should be interpreted as the numerically best model in the present controlled dataset, rather than as a universally superior model. The broader conclusion is that compact statistical descriptors provide a strong and practical route for target-tool adaptation, especially when combined with tree-based ensemble regressors and low-label affine calibration.

The hierarchical signal statistical fusion model performs competitively but does not uniformly outperform the strongest feature-based baseline. Under pass-uniform calibration with $N_{\text{pass}} = 30$, the fusion model obtains an RMSE of 0.0949 μm and an R^2 of 0.4357, which is near RF and better than SVR-RBF, SVR-Linear, and GPR. Under the stricter layer-one-pass protocol, the fusion model achieves an RMSE of 0.1046 μm and an R^2 of 0.1880. These results suggest that the CNN waveform branch and masked layer aggregation do not replace the importance of statistical descriptors. Instead, the fusion model should be interpreted as a structured signal statistical alternative that preserves the pass-layer data organization while providing competitive low-label target-tool adaptation.

Overall, Table 3 shows that raw cross-tool prediction remains unreliable, whereas affine calibration substantially improves all compared models. RF achieves the best numerically calibrated performance on the present dataset. However, the modest margins over LSBoost and the fusion comparison model indicate that the main conclusion should be the effectiveness of statistical descriptors and low-label affine calibration, rather than the universal superiority of a single regressor.

4.3 Raw Prediction and Pass-Uniform Calibration Analysis of the RF Model

Since RF achieves the best numerically calibrated performance among the compared models in the present dataset, it is selected for detailed visualization of raw and pass-uniform calibrated predictions in Fig. 9. This selection demonstrates the calibration behavior of the strongest numerical baseline without implying the universal superiority of RF. Panels (a), (c), and (e) show the raw cross-tool predictions for Fold 1–Fold 3, respectively. Panels (b), (d), and (f) show the corresponding pass-uniform affine-calibrated results using $N_{\text{pass}} = 30$ target-tool samples.

The raw prediction results exhibit clear target-dependent mismatch. For Fold 1 targeting C1, the raw RMSE is 0.2166 μm and the R^2 is -2.5904 ; for Fold 2 targeting C2, 0.3795 μm and -8.7477 ; and for Fold 3 targeting C3, 0.3675 μm and -3.4897 . These negative R^2 values indicate

that directly applying a source-trained model to a new tool is worse than using the target-tool mean predictor. Therefore, raw cross-tool prediction remains unreliable even when all 72 statistical descriptors and per-tool unsupervised normalization are used.

The RMSE values decrease to 0.0899 μm , 0.1062 μm , and 0.0815 μm for Fold 1, Fold 2, and Fold 3, respectively. The corresponding R^2 values become positive, reaching 0.3817, 0.2363, and 0.7793, respectively. On average, pass-uniform calibration reduces the RF RMSE from 0.3212 μm to 0.0925 μm and improves the average R^2 from -4.9426 to 0.4658. This comparison confirms that the dominant cross-tool error is not purely random scatter; instead, it contains systematic output bias and scale mismatch that can be effectively corrected using a few labeled target-tool samples.

Notably, that pass-uniform calibration is interpreted as a controlled full-life adaptation protocol rather than a strictly online early-stage deployment setting. Since the calibration samples are uniformly selected across the entire target-tool life, this protocol models the potential performance when limited but representative target-tool labels are available. Therefore, stricter layer-wise calibration protocols are further analyzed in the following subsection to provide a more conservative evaluation of low-label target-tool adaptation [44].

4.4 Layer-Wise Calibration Protocol Comparison

Although pass-uniform calibration provides the best average RF performance, it should be considered a controlled full-life adaptation setting. In practical deployment, calibration labels uniformly spanning the entire target-tool life may not always be available at the initial stage of a new tool. Therefore, two layer-wise calibration protocols are further evaluated, and the corresponding prediction-error distributions are compared in Fig. 10.

The layer-uniform all-pass protocol selects $N_{\text{layer}} = 10$ target-tool layers uniformly across tool life and uses all valid passes in these selected layers for affine calibration. Under this setting, the RF model achieves an average RMSE of 0.0956 μm and an average R^2 of 0.3331. The layer-all-pass protocol shows slightly weaker performance than the pass-uniform result of 0.0925 μm and 0.4658, but it still provides a substantial improvement over raw cross-tool prediction. This comparison indicates that representative layer-level calibration samples can effectively correct part of the output bias and scale mismatch caused by tool-to-tool shift.

The layer-uniform one-pass protocol is stricter. In this setting, only one valid pass from each selected calibration layer is used to fit the affine mapping, and these layers are excluded from the final evaluation set. This design reduces the possibility that within-layer pass correlation leads to

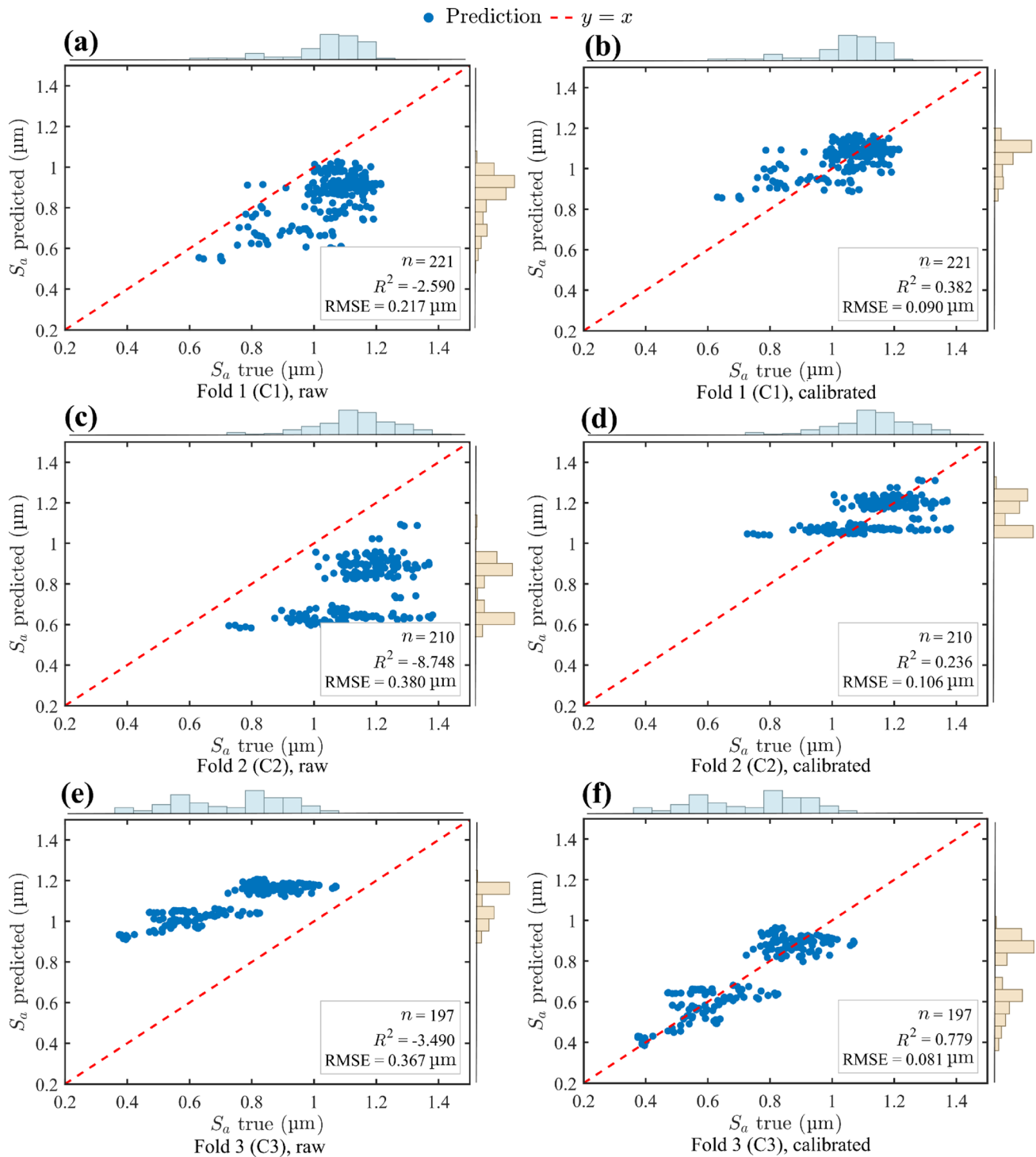


Fig. 9 Scatter plot comparison of raw and pass-uniform calibrated predictions of the RF model under the LOTO cross-tool evaluation. Panels (a), (c), and (e) show the raw prediction results for Fold 1–Fold 3, respectively, whereas panels (b), (d), and (f) show the cor-

responding pass-uniform calibrated results with $N_{\text{pass}} = 30$. For fair comparison, raw and calibrated results are evaluated on the same target-tool evaluation subset after the samples used for affine calibration are excluded

overly optimistic results. Under this protocol, RF achieves an average RMSE of $0.0973 \mu\text{m}$ and an average R^2 of 0.2908 . Although the performance is slightly lower than that of

layer-all-pass calibration, the result remains near the other calibrated protocols and is far better than the raw prediction. This comparison demonstrates that even sparse layer-wise

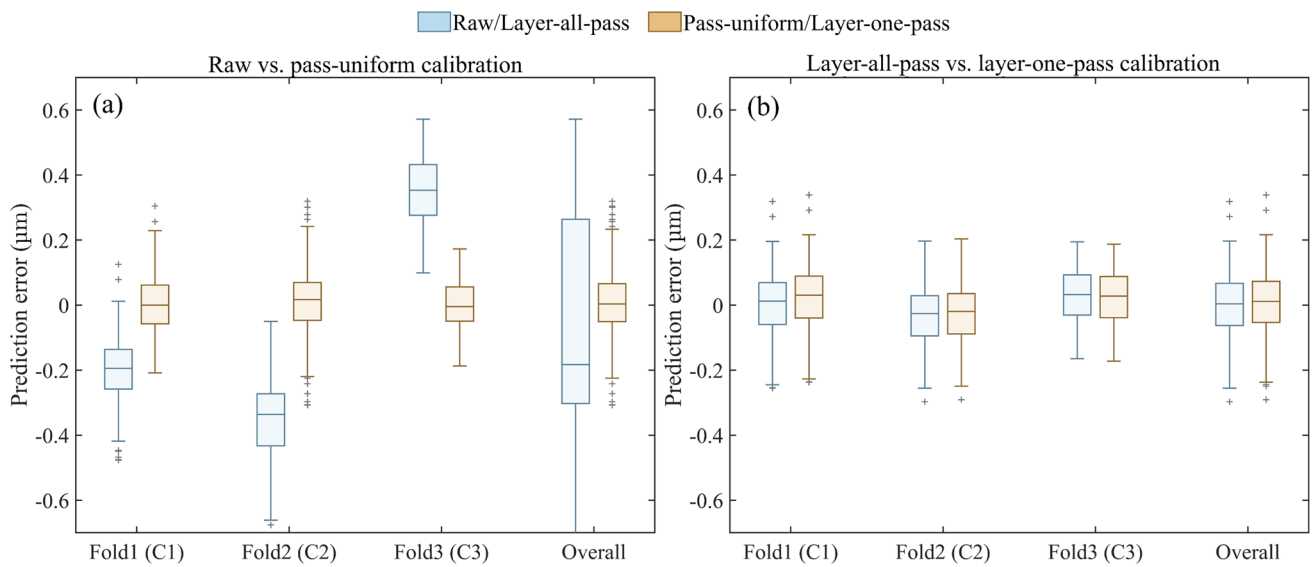


Fig. 10 Boxplot comparison of prediction errors of the RF model under different calibration protocols. Panel (a) compares raw prediction and pass-uniform affine calibration with $N_{\text{pass}} = 30$, whereas

panel (b) compares layer-uniform all-pass and layer-uniform one-pass calibration with $N_{\text{layer}} = 10$. The error is defined as the predicted S_a minus the measured S_a

target-tool labels can provide useful information for output-level correction.

The fold-wise results also reveal target-dependent differences. For Fold 3 targeting C3, both layer-wise protocols yield high positive R^2 values, with 0.7223 for layer-all-pass and 0.7327 for layer-one-pass. In contrast, Fold 1 targeting C1 remains more challenging under the stricter layer-one-pass protocol, whose R^2 is -0.1008 despite the RMSE being reduced to $0.1102 \mu\text{m}$. This finding suggests that calibration can substantially reduce absolute error, but the ability to recover target-tool variation still depends on the roughness distribution and fold-specific response characteristics.

Overall, Fig. 10 shows that the prediction error distributions become more concentrated around zero after affine calibration. The reductions in mean error and error dispersion support the interpretation that few-shot affine calibration mainly corrects systematic output bias and scale mismatch. Therefore, the key deployment implication is that raw source-trained prediction should not be directly trusted after tool replacement, whereas a few target-tool roughness measurements can markedly improve the reliability of cross-tool S_a prediction.

4.5 Discussion and Limitations

The proposed framework should be distinguished from conventional transfer learning or domain adaptation methods. Its main purpose is not to learn a generic domain-invariant representation but to provide a practical measurement adaptation procedure after tool replacement. In this procedure, target-tool unlabeled descriptors are first

used to reduce input-scale mismatch, and a few measured target-tool S_a values are then used to correct output bias through affine calibration. This separation between unsupervised input alignment and low-label output correction is consistent with the practical metrology workflow in which cutting signals can be continuously collected, whereas surface topography measurements are available only for a few target-tool samples.

The results further support the rationale of this two-stage adaptation design. Per-tool normalization and affine calibration address two types of tool-replacement-induced mismatch. Per-tool normalization uses unlabeled target-tool descriptors to reduce differences in input-level scale and distribution. Conversely, affine calibration uses a few measured target-tool S_a values to correct output-level offset and scale mismatch. The negative raw R^2 values in Table 3 indicate that input-level alignment alone is insufficient for reliable target-tool prediction. In contrast, the improved calibrated results in Table 3 and Fig. 9 show that a simple affine mapping can effectively correct the remaining dominant error component. Therefore, the proposed framework is intentionally designed as a lightweight measurement adaptation procedure rather than a complex target-domain retraining strategy, which would be more prone to overfitting across limited target-tool S_a labels.

The results indicate that S_a prediction after tool replacement is mainly limited by target-tool output bias and scale mismatch. Although per-tool unsupervised normalization reduces input-level descriptor differences, all compared models still produce negative average R^2 values before affine calibration. Therefore, raw source-trained prediction should

not be directly trusted as a reliable estimate of the measured S_a value after a new tool is installed [20, 41].

A key implication for surface quality metrology is that compact statistical descriptors provide a robust representation of the cutting state under the present small-sample setting. In the present controlled three-tool LOTO setting, RF with all 72 descriptors achieves the best numerically calibrated performance. However, the performance margins between RF, LSBoost, and the hierarchical signal statistical fusion model are modest under some calibrated protocols. For example, under pass-uniform calibration, RF, LSBoost, and the fusion model obtain RMSEs of 0.0925 μm , 0.0952 μm , and 0.0949 μm , respectively. Therefore, this result should not be interpreted as universal RF dominance. Instead, it suggests that compact statistical descriptors provide a reliable and practical route for target-tool adaptation, especially when combined with tree-based ensemble regressors and low-label affine calibration. The fusion model remains a meaningful structured comparison model. However, with the present three-tool small-sample dataset, a more complex waveform fusion architecture may not consistently outperform compact descriptor-based regressors [44].

Moreover, the calibration protocol affects the interpretation of the results. Pass-uniform calibration provides the best average performance because the selected calibration samples span the entire life of the target tool. Therefore, it should be considered a controlled full-life adaptation upper bound rather than a strict early-stage online deployment condition. This result should not be interpreted as the performance achievable immediately after installing a new tool, because uniformly distributed labels over the full target-tool life are generally unavailable at the beginning of machining. The layer-wise protocols provide more conservative evaluations. In particular, the layer-one-pass protocol uses only one valid pass from each selected layer, excluding the selected calibration layers from final evaluation. This approach reduces the influence of within-layer pass correlation [44]. Therefore, the three protocols should be considered complementary evaluation settings: pass-uniform gives an idealized upper-bound estimate, layer-all-pass represents representative stage-wise inspection, and layer-one-pass provides a stricter sparse-label assessment.

The role of per-tool normalization should also be clearly defined. In this study, per-tool normalization uses only unlabeled target-tool signal/statistical feature distributions and does not use target-tool S_a labels, so it is not label leakage. Specifically, the target-tool normalization statistics are estimated using all valid unlabeled passes collected over the target-tool life. Therefore, the present raw setting should be interpreted as an offline unsupervised input-level alignment setting, rather than a strict source-only transfer or early-stage online deployment. This setting should be interpreted as cross-tool prediction with unsupervised input-level

alignment, followed by limited-label affine calibration [41, 44]. This distinction is important for avoiding overly optimistic claims about source-only generalization.

From an industrial-use perspective, the proposed framework can be interpreted as a measurement-assisted adaptation procedure after tool replacement. First, a source prediction model can be trained on historical labeled data from previously used tools, where each valid pass-level force/vibration sample is paired with a measured S_a value. Second, after a new target tool is installed, force/vibration signals can be collected during normal cutting, and the resulting unlabeled target-tool descriptors can be used to estimate per-tool normalization statistics for input-level alignment. Third, a few target-tool passes, or representative layers, can be selected for offline surface topography measurement. These limited measured values of target-tool S_a are then used to fit the affine calibration parameters. Finally, the calibrated model can be applied to the remaining target-tool passes to provide signal-based S_a estimates. In this workflow, dense roughness measurements of every target-tool pass are unnecessary, while limited metrology feedback is still used to correct target-tool output bias and improve prediction reliability.

Several limitations remain. First, the dataset contains only three tools under a single fixed cutting parameter setting. This design enables a strict, three-fold LOTO evaluation in which each tool is held out once as an unseen target tool, and the remaining two tools are used as source tools. Therefore, the current dataset is suitable for examining whether low-label target-tool adaptation is feasible under controlled cutting conditions and for isolating tool-replacement-induced shift from cutting-parameter variation [20]. However, three tools cannot support broad generalization claims across different machines, tool batches, cutting parameters, materials, or industrial production environments. Consequently, the present results should be interpreted as evidence of feasibility within the current controlled experimental setting rather than a complete proof of universal tool-replacement generalization. Future work should include more tools, repeated tool replacements, different tool batches, and broader cutting conditions to further evaluate the robustness and industrial generalizability of the proposed framework.

Second, the calibration protocols still require limited target-tool S_a measurements. In particular, pass-uniform calibration uses target-tool labels distributed throughout the available tool life and may be unavailable at the early stage of a new tool life. Therefore, it should be interpreted as a controlled full-life adaptation upper-bound protocol rather than a strictly online early-stage deployment protocol [44]. The layer-all-pass and layer-one-pass protocols offer more conservative layer-wise calibration scenarios, but they still depend on target-tool surface topography measurements. Future work should further investigate early-life calibration strategies with smaller target label budgets and more realistic

online measurement schedules. For practical deployment, the layer-wise protocols and future early-life calibration strategies are more relevant than the pass-uniform upper-bound protocol because they require fewer, more localized target-tool measurements.

Third, the current per-tool normalization uses the available unlabeled target-tool descriptor distribution in an offline adaptation manner. This setting is useful for evaluating whether unlabeled target-tool signals reduce input-level descriptor mismatch, but it should not be interpreted as strictly source-only or fully online. Future online deployment should further investigate rolling or incremental normalization strategies, in which target-tool normalization statistics are updated progressively as new unlabeled cutting signals are collected [41].

Fourth, tool wear is physically related to surface generation, but the final selected prediction model does not use VB or wear stage labels as supervised inputs because S_a is measured at the pass level, whereas VB is measured at the layer level. Thus, VB should be interpreted as an experimental tool-life characterization variable and a mechanism interpretation variable in this study, rather than as an online input variable of the final prediction model. This choice also reflects the intended deployment scenario, in which force/vibration signals and limited S_a measurements are more readily available for target-tool adaptation than frequent direct VB measurements.

Finally, although three repeated surface topography measurements were averaged to obtain each pass-level S_a label, the repeated-measurement uncertainty was not independently quantified as a separate uncertainty component. Therefore, the reported prediction errors may still include model prediction error and residual surface measurement variability. Future work should further quantify the metrology uncertainty in repeated S_a measurements and separate it from model prediction error.

5 Conclusions

This study investigated measurement-driven low-label target-tool adaptation for signal-based S_a prediction in Ti-6Al-4V milling. A one-pass-one-sample dataset was constructed from steady-state force/vibration signals, and each valid pass was associated with one measured S_a value obtained from surface topography measurement [30, 31]. Seventy-two statistical descriptors were retained to characterize the cutting state. Descriptor-based regressors and a hierarchical signal statistical fusion model were compared under identical three-fold LOTO protocols, per-tool unsupervised normalizations, and fixed-budget affine calibration protocols [41, 44]. The main conclusions are as follows:

1. Direct raw cross-tool S_a prediction remains unreliable after tool replacement. All compared models produce negative average R^2 values before affine calibration, indicating that input-level alignment and model representation alone cannot fully remove target-tool output bias and scale mismatch.
2. Few-shot affine calibration substantially improves the reliability of target-tool S_a prediction. By fitting a simple affine mapping with a limited number of measured target-tool S_a samples, prediction errors are reduced under pass and layer-wise calibration protocols.
3. In the present controlled three-tool LOTO setting, RF with all 72 statistical descriptors achieves the best numerically calibrated performance. Its average RMSE/ R^2 values are 0.0925 μm /0.4658 under pass-uniform calibration, 0.0956 μm /0.3331 under layer-all-pass calibration, and 0.0973 μm /0.2908 under layer-one-pass calibration. However, the margins over LSBoost and the fusion comparison model are modest under some protocols. Therefore, this result should not be interpreted as universal RF dominance. Instead, it suggests that compact statistical descriptors offer a reliable and practical route for low-label target-tool adaptation, especially when combined with tree-based ensemble regressors and few-shot affine calibration.
4. The hierarchical signal statistical fusion model remains competitive but does not consistently outperform RF. This comparison suggests that, under the current three-tool small-sample setting, reliable statistical descriptors and affine output correction are more important than increasing model complexity.
5. The proposed framework clarifies the boundary between strict source-only prediction and practical low-label target-tool adaptation. Per-tool normalization should be interpreted as unsupervised input-level alignment, whereas affine calibration provides output-level correction using a limited number of measured target-tool S_a values.

Overall, the most reliable approach in the present controlled Ti-6Al-4V milling dataset is to combine informative statistical descriptors, robust descriptor-based regression, and few-shot affine calibration. These results demonstrate the feasibility of the proposed low-label target-tool adaptation framework under the current three-tool LOTO setting, but further validation with more tools, different tool batches, and broader cutting conditions remains necessary before broad industrial generalization can be claimed. Future work will expand the dataset to include more tools and cutting conditions and further investigate early-life calibration strategies for online deployment after tool replacement.

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Data Availability The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Code Availability The code used to support the findings of this study is available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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